ECA5801-IOE AND AI DRIVEN SYSTEMS

EXPERIMENT 4

SIMULATION

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4. Simulation and Comparative Performance Evaluation of Zigbee, 6LoWPAN and NB-IoT Communication Protocols for IoE Applications using NS-3

CODE:

```
# ================================================================
# SIMULATION AND COMPARATIVE PERFORMANCE EVALUATION
# OF Zigbee, 6LoWPAN AND NB-IoT
# Python-Based IoE Simulation
# ================================================================

# ------------------------------
# 1. IMPORT LIBRARIES
# ------------------------------

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# ------------------------------
# 2. DEFINE PROTOCOLS
# ------------------------------

protocols = [
    "Zigbee",
    "6LoWPAN",
    "NB-IoT"
]

# ------------------------------
# 3. SIMULATION PARAMETERS
# ------------------------------

num_packets = 5000
packet_size = 200          # bytes
simulation_time = 10       # seconds

# ------------------------------
# 4. PROTOCOL PARAMETERS
```

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# ------------------------------

# Data rate in kbps
data_rate = {
    "Zigbee": 250,
    "6LoWPAN": 200,
    "NB-IoT": 100
}

# Base delay in milliseconds
base_delay = {
    "Zigbee": 20,
    "6LoWPAN": 30,
    "NB-IoT": 50
}

# Packet loss probability
packet_loss = {
    "Zigbee": 0.02,
    "6LoWPAN": 0.015,
    "NB-IoT": 0.01
}

# Transmission power in watts
tx_power = {
    "Zigbee": 0.05,
    "6LoWPAN": 0.06,
    "NB-IoT": 0.20
}

# Reception power in watts
rx_power = {
    "Zigbee": 0.03,
    "6LoWPAN": 0.035,
    "NB-IoT": 0.05
}

# ------------------------------
# 5. START SIMULATION
# ------------------------------

np.random.seed(42)

results = []

for protocol in protocols:

    # ========================================
```

```
# PACKET TRANSMISSION
# ========================================

sent_packets = num_packets

# Simulate packet loss
lost_packets = np.random.binomial(
    sent_packets,
    packet_loss[protocol]
)

received_packets = (
    sent_packets - lost_packets
)

# ========================================
# PACKET DELIVERY RATIO
# ========================================

pdr = (
    received_packets /
    sent_packets
) * 100

# ========================================
# THROUGHPUT
# ========================================

# Offered traffic load
offered_load = (
    sent_packets *
    packet_size *
    8 /
    simulation_time /
    1000
)

# Maximum capacity of protocol
protocol_capacity = data_rate[protocol]

# Actual throughput
throughput = min(
    offered_load * (pdr / 100),
    protocol_capacity
)

# ========================================
# DELAY
```

```
# ==========================================

# Traffic congestion effect
congestion_factor = (
    offered_load /
    protocol_capacity
)

average_delay = (
    base_delay[protocol] *
    (1 + 0.3 * min(congestion_factor, 3))
)

# Small random variation
average_delay += np.random.uniform(-2, 2)

average_delay = max(
    average_delay,
    1
)

# ==========================================
# ENERGY CONSUMPTION
# ==========================================

actual_bits = (
    throughput *
    1000 *
    simulation_time
)

transmission_time = (
    actual_bits /
    (protocol_capacity * 1000)
)

# Transmission energy
tx_energy = (
    tx_power[protocol] *
    transmission_time
)

# Reception energy
rx_energy = (
    rx_power[protocol] *
    transmission_time
)
```

```
    # Total energy
    total_energy = (
        tx_energy +
        rx_energy
    )

    # ========================================
    # STORE RESULTS
    # ========================================

    results.append({
        "Protocol": protocol,
        "Throughput (kbps)": throughput,
        "Delay (ms)": average_delay,
        "PDR (%)": pdr,
        "Energy (J)": total_energy
    })


# ============================================================
# 6. CREATE RESULT TABLE
# ============================================================

df = pd.DataFrame(results)

df = df.round({
    "Throughput (kbps)": 2,
    "Delay (ms)": 2,
    "PDR (%)": 2,
    "Energy (J)": 2
})


# ============================================================
# 7. PRINT RESULT TABLE
# ============================================================

print("\n================================================")
print("       IoE PROTOCOL PERFORMANCE RESULTS")
print("================================================\n")

display(df)


# ============================================================
# 8. CREATE FOUR COMPACT GRAPHS
# ============================================================

fig, axes = plt.subplots(
```

```
    2, 2,
    figsize=(7, 4.5)
)


# ---------- Throughput ----------

axes[0, 0].bar(
    df["Protocol"],
    df["Throughput (kbps)"]
)

axes[0, 0].set_title(
    "Throughput",
    fontsize=10
)

axes[0, 0].set_ylabel(
    "kbps",
    fontsize=8
)

axes[0, 0].tick_params(
    labelsize=7
)


# ---------- Delay ----------

axes[0, 1].bar(
    df["Protocol"],
    df["Delay (ms)"]
)

axes[0, 1].set_title(
    "Delay",
    fontsize=10
)

axes[0, 1].set_ylabel(
    "ms",
    fontsize=8
)

axes[0, 1].tick_params(
    labelsize=7
)
```

```
# ---------- PDR ----------

axes[1, 0].bar(
    df["Protocol"],
    df["PDR (%)"]
)

axes[1, 0].set_title(
    "Packet Delivery Ratio",
    fontsize=10
)

axes[1, 0].set_ylabel(
    "%",
    fontsize=8
)

axes[1, 0].set_ylim(
    0,
    100
)

axes[1, 0].tick_params(
    labelsize=7
)


# ---------- Energy ----------

axes[1, 1].bar(
    df["Protocol"],
    df["Energy (J)"]
)

axes[1, 1].set_title(
    "Energy Consumption",
    fontsize=10
)

axes[1, 1].set_ylabel(
    "J",
    fontsize=8
)

axes[1, 1].tick_params(
    labelsize=7
)
```

```
plt.tight_layout(
    pad=1
)

plt.show()


# ============================================================
# 9. PERFORMANCE COMPARISON
# ============================================================

best_throughput = df.loc[
    df["Throughput (kbps)"].idxmax(),
    "Protocol"
]

lowest_delay = df.loc[
    df["Delay (ms)"].idxmin(),
    "Protocol"
]

best_pdr = df.loc[
    df["PDR (%)"].idxmax(),
    "Protocol"
]

lowest_energy = df.loc[
    df["Energy (J)"].idxmin(),
    "Protocol"
]


print("\n================================================")
print("             COMPARISON RESULT")
print("================================================")

print(
    "Highest Throughput :",
    best_throughput
)

print(
    "Lowest Delay       :",
    lowest_delay
)

print(
    "Highest PDR        :",
    best_pdr
```

```
)

print(
    "Lowest Energy      :",
    lowest_energy
)


# ============================================================
# 10. FINAL CONCLUSION
# ============================================================

print("\n================================================")
print("                 CONCLUSION")
print("================================================")

print(
    f"{best_throughput} provides the highest throughput."
)

print(
    f"{lowest_delay} provides the lowest delay."
)

print(
    f"{best_pdr} provides the highest packet delivery ratio."
)

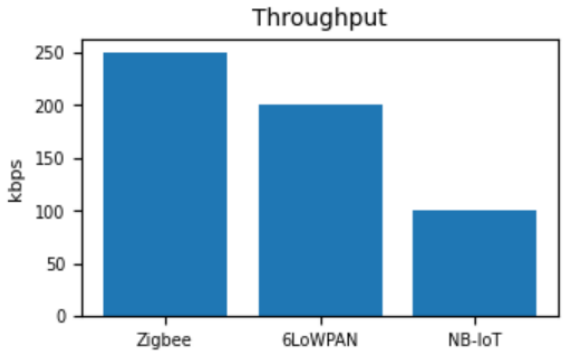
print(
    f"{lowest_energy} consumes the least energy."
)

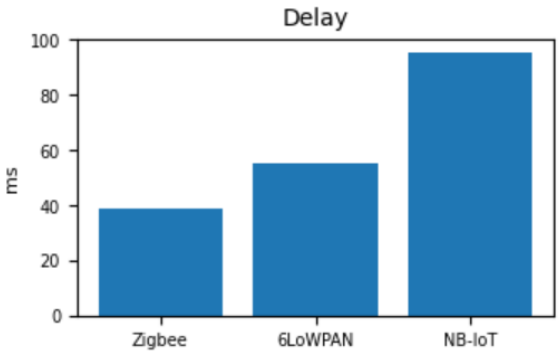
print(
    "\nThe simulation shows that Zigbee, 6LoWPAN and NB-IoT "
    "have different performance characteristics in terms of "
    "throughput, delay, packet delivery ratio and energy "
    "consumption."
)
```

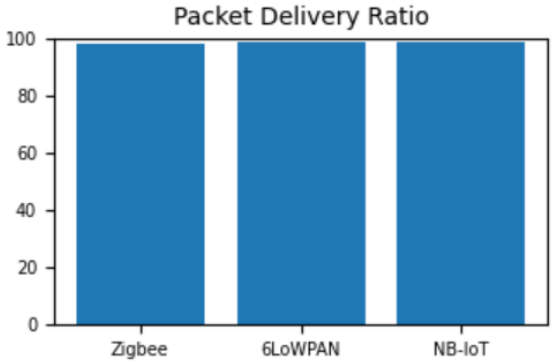
OUTPUT:

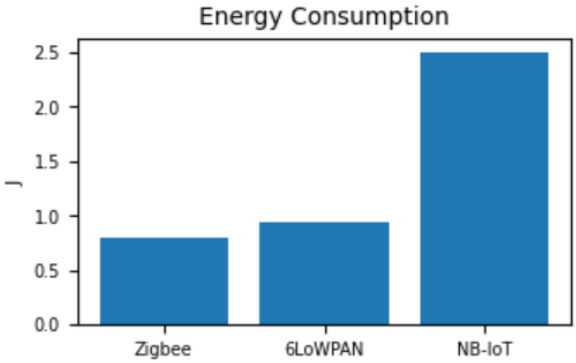
```
================================================
       IoE PROTOCOL PERFORMANCE RESULTS
================================================
```

	Protocol	Throughput (kbps)	Delay (ms)	PDR (%)	Energy (J)
0	Zigbee	250	38.93	98.10	0.80
1	6LoWPAN	200	55.62	98.64	0.95
2	NB-IoT	100	95.40	99.16	2.50










```
================================================
              COMPARISON RESULT
================================================
Highest Throughput : Zigbee
Lowest Delay       : Zigbee
Highest PDR        : NB-IoT
Lowest Energy      : Zigbee

================================================
                  CONCLUSION
================================================
Zigbee provides the highest throughput.
Zigbee provides the lowest delay.
NB-IoT provides the highest packet delivery ratio.
Zigbee consumes the least energy.
```